BRIAN KIPRUTO KOSGEI.

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SUMMARY

A highly motivated **Nuclear & computational physicist** with a strong foundation in computational methods and a passion for building systems to solve complex scientific problems for the safe and secure development of nuclear technology in Africa. Possesses a proven track record of designing, building, and deploying **end-to-end solutions**, from robotic platforms to scalable web applications.

A SNAPSHOT OF MY APPLIED RESEARCH & INNOVATION.

1. **Public Speaking & Research Dissemination:** Secured oral presentation slots for Project R.A.N.E.G.R. at the African Youth Nuclear Summit (AYNS2025) and the UoN Annual International Engineering Conference (EC2025), showcasing the ability to present complex research to professional audiences.
2. **Content & Thought Leadership:** I Author and manage a personal blog, 'A Physicist's Notebook,' to document research and make complex scientific concepts accessible to the public, demonstrating a proactive approach to knowledge transfer.
3. **Media Recognition:** Featured in the **African Physics Newsletter** and **Citizen Digital** for work in developing innovative STEM curricula, thus showcasing my ability to generate and promote impactful scientific stories.
4. **Educational Outreach:** Served as a STEM Educator at The Bytelab, where I designed and delivered curriculum in electronics, coding, and robotics for young learners. This experience honed my ability to communicate technical subjects effectively to diverse audiences.

WORK EXPERIENCE

Autonomous Radiation Mapping Robot - ByteAnza Research — Jan 2025-present

https://byteanza.com/research/portfolio-details.html

Led the development of a robotic platform for democratizing environmental data, showcasing an innovative, youth-led African solution in AI and robotics." This still shows your technical leadership but with a focus on impact..

- The system integrates a 6WD robotic platform with a micro-controller, On-Board Computer, Geiger-Müller tube, GPS, and IMU to acquire precise spatial measurements of radiation levels.
- Developed software for data processing and map generation, enabling a deeper understanding of radiation dispersion and its implications for environmental and public health, and providing valuable insights for developing effective monitoring and mitigation strategies for radiological hazards.

STEM Educator, The Bytelab. (Part time) — Nov 2023 - Present

http://www.byteanza.com/bytelab/

The ByteLab aims to foster a love for STEM in children from a young age through hands-on Experiential Learning.

- Led the planning, development, and execution of dynamic STEM education programs for young learners, encompassing electronics, coding, robotics, and IoT.
- Managed program delivery, including instructional sessions on microcontroller programming (Arduino), circuit design, sensor integration, and automation, guiding students through hands-on project development and prototyping.
- Developed scientific and technological curriculum and learning resources, ensuring engaging and effective content delivery.
- Mentored students, providing career guidance and fostering interest in STEM fields.

Scientific Researcher, ByteAnza. — Jan 2023 - Nov 2023

www.Byteanza.com

- Led the research, development, and implementation of Bytegrow, an innovative project delivering smart, autonomous agricultural solutions for small- and medium-scale farmers.
- Managed the project lifecycle, from prototype development (a fogponics system for water conservation using Arduino and ESP8266 microcontrollers) to data collection and cloud storage integration.
- Achieved project milestones, resulting in recognition as a top 10 finalist in the AYUTE Kenya Challenge 2023 and a winning entry at the KNBS Hackathon 2023.

Machine Learning Research Intern & Project team member, University of Konstanz. Germany. — **May 2022 - Aug 2022**

- Contributed to a research project focused on machine learning applications for particulate structure detection in colloidal solutions.
- Conducted data analytics and visualization using Ovito.
- Participated in the development and implementation of neural networks for studying and predicting particulate structures.

EDUCATION

MSc Nuclear Science & Technology — **Aug 2025 - Present**

University of Nairobi.- Institute of Nuclear Science & Technology
My research revolves around using machine learning to build a virtual prototyping tool for rapid performance prediction of flexible Graded-Z shielding material composites.

BSc Astrophysics — **Sep 2019 - Sep 2023**

University of Nairobi - Department of Physics

- Second Class Honours.

VOLUNTEER WORK

Nuclear Society of Kenya (Member) — **Jan 2025 - Present**

My volunteer work with the Nuclear Society of Kenya allows me to engage in public outreach, professional development, and policy advocacy, which aligns directly with my long-term goal of fostering a strong and responsible nuclear ecosystem in Africa. This experience further motivates my pursuit of advanced studies to acquire the expertise necessary to lead in this crucial area.

Volunteer Researcher, International Astronomical Asteroid search campaign — **Oct 2021 - Nov 2021**

- Applied data analysis techniques to telescope data to identify and locate potential asteroids using Astrometrica software.

Moon Village Association — **Mar 2021 - Nov 2022**

- Served as Deputy Coordinator at MVA-PESC Kenya (Moon Village Association - Participation of Emerging Space Countries) in conjunction with the Kenya Space Agency.

FEATURES & CERTIFICATIONS

The African Physics Newsletter.
Website: https://www.aps.org/publications/african-physics-newsletter

Got Featured in the July 2024 issue of the African Physics Newsletter for my works at the Bytelab. Where curiosity meets innovation.

Certified Space Professional, Teaching Science & Technology, Inc. (TSTI)
Website: https://www.tsti.net/

The Space Mission Fundamentals Course by TSTI Inc., in collaboration with the Kenya Space Agency, provides a comprehensive overview of space systems, missions, and operations. It covers key topics such as orbital mechanics, space environment, mission management, spacecraft subsystems, launch vehicles, and human spaceflight challenges. Designed for engineers, scientists, and professionals new to the space industry, the course equips learners with the ability to analyze space missions, understand trade-offs in system design, and apply space concepts to real-world problems. Through practical exercises, participants gain hands-on experience in mission planning and spacecraft design.

Qubit by Qubit's Introduction to Quantum Computing

Website:_ gg@the-cs.org

An introductory course in quantum computing. The aim is to expose students to this emerging field while providing them with real-world programming skills for quantum computers. In the course, We learnt about the foundational concepts in quantum computation, including superposition, entanglement, superdense coding, and explored quantum applications.

Citizen Digital

https://www.citizen.digital/tech/the-bytelab-where-curiosity-meets-innovation-n361062

“The future of technological innovation lies at the intersection of intelligent systems, sustainability, and global collaboration.....”

LANGUAGE SKILLS

Mother tongue(s): Swahili

English- IELTS

LISTENING: 8.5 READING: 8.0 WRITING: 7.0 SPEAKING: 7.5 Overall : 8.0 (C1)

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

German

Horen (Listening**) A2** **Lasen(**reading**) A2**
Schreiben(Writing)A2 **Sprechen** (speaking) A2

Levels: A1 and A2: Basic & Elementary level; B1 and B2: Intermediate & Upper intermediate; C1 and C2: Advanced & near-native level.

TECHNICAL SKILLS

- **Programming & Scripting:** Python, C++, JavaScript, Arduino
- **AI & Machine Learning:** Machine Learning (ML), AI Integration (Google Gemini API), Gradient Boosting Machines, Data Analysis & Visualization
- **Backend & Frameworks:** Django, Django Channels, Celery, Redis
- **Databases:** PostgreSQL, MySQL/MariaDB, Django ORM
- **Frontend:** HTML5, CSS3, Vanilla JavaScript, React.js
- **Scientific & Simulation:** GEANT4, Monte Carlo Methods
- **Hardware & IoT:** Raspberry Pi, Arduino, Sensor Integration (GPS, IMU, Geiger Counters, PM Sensors etc.)
- **Infrastructure & Ops:** Caching (Memcached), Background Tasks (Celery Beat), REST APIs, WebSockets

OTHER SKILLS

- **Research & Analysis:** Physics Research, Qualitative & Quantitative analysis
- **Regulatory Project management:** Project Management, Strategic Communication, Regulatory Compliance.